“Supplemental SOW Documents”

10/2/2026

Solicitation: 47PD5526Q0051

Project: FHWA Renovations

Additional Documents for:

1. Roller Shade Specifications
2. Jogged Wall Detail
3. STC 45 Wall Detail

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SECTION 122413 - ROLLER WINDOW SHADES

TIPS:

To view non-printing **Editor's Notes** that provide guidance for editing, click on Masterworks/Single-File Formatting/Toggle/Editor's Notes.

To read **detailed research, technical information about products and materials, and coordination checklists**, click on Masterworks/Supporting Information.

Revise this Section by deleting and inserting text to meet Project-specific requirements.

This Section uses the term "Architect." Change this term to match that used to identify the design professional as defined in the General and Supplementary Conditions.

Verify that Section titles referenced in this Section are correct for this Project's Specifications; Section titles may have changed.

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

Retain or delete this article in all Sections of Project Manual.

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Manually operated roller shades with [**single**] [**double**] rollers.
2. Manually operated roller shades for skylights.
3. Motor-operated roller shades with [**single**] [**double**] rollers.
4. Motor-operated roller shades for skylights.

B. Related Requirements:

Retain subparagraphs below to cross-reference requirements Contractor might expect to find in this Section but are specified in other Sections.

1. Section [**061000 "Rough Carpentry"**] [**061053 "Miscellaneous Rough Carpentry"**] for wood blocking and grounds for mounting roller shades and accessories.
2. Section 079200 "Joint Sealants" for sealing the perimeters of installation accessories for light-blocking shades with a sealant.

1.3 ALLOWANCES

Retain products and Work included in this Section that are covered by cash or quantity allowance. Do not include amounts. Insert descriptions of items in Part 2 or 3 to provide information affecting the cost of the Work that is not included under the allowance.

A. Roller shades are part of [**Window-Covering Allowance**] <**Insert name of allowance**>.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, features, finishes, and operating instructions for roller shades.

B. Shop Drawings: Show fabrication and installation details for roller shades, including shadeband materials, their orientation to rollers, and their seam and batten locations.

1. Motor-Operated Shades: Include details of installation and diagrams for power, signal, and control wiring.

Retain "Samples" Paragraph below for single-stage Samples, with a subordinate list if applicable. Retain "Samples for Initial Selection" and "Samples for Verification" paragraphs for two-stage Samples.

C. Samples: For each exposed product and for each color and texture specified, 10 inches (250 mm) long.

D. Samples for Initial Selection: For each type and color of shadeband material.

1. Include Samples of accessories involving color selection.

E. Samples for Verification: For each type of roller shade.

1. Shadeband Material: Not less than [**10 inches (250 mm)**] [**3 inches (76 mm)**] square. Mark interior face of material if applicable.
2. Roller Shade: Full-size operating unit, not less than 16 inches (400 mm) wide by 36 inches (900 mm) long for each type of roller shade indicated.
3. Installation Accessories: Full-size unit, not less than 10 inches (250 mm) long.

F. Product Schedule: For roller shades.[**Use same designations indicated on Drawings.**]

1.5 INFORMATIONAL SUBMITTALS

Coordinate "Qualification Data" Paragraph below with qualification requirements in Section 014000 "Quality Requirements" and as may be supplemented in "Quality Assurance" Article.

A. Qualification Data: For Installer.

Retain "Product Certificates" Paragraph below to require submittal of product certificates from manufacturers.

B. Product Certificates: For each type of shadeband material.

Retain "Product Test Reports" Paragraph below to require submittal of product test reports with or without product certificates from manufacturers.

C. Product Test Reports: For each type of shadeband material, for tests performed by [**manufacturer and witnessed by a qualified testing agency**] [**a qualified testing agency**].

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For roller shades to include in maintenance manuals.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Roller Shades: Full-size units equal to 5 percent of quantity installed for each size, color, and shadeband material indicated, but no fewer than [**two**] <**Insert number**> units.

1.8 QUALITY ASSURANCE

A. Installer Qualifications: Fabricator of products.

B. Mockups: Build mockups to verify selections made under Sample submittals, to demonstrate aesthetic effects, and to set quality standards for fabrication and installation.

1. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.

Retain subparagraph below if the intention is to make an exception to the default requirement in Section 014000 "Quality Requirements" for demolishing and removing mockups.

2. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Deliver roller shades in factory packages, marked with manufacturer, product name, and location of installation using same designations indicated on Drawings.

1.10 FIELD CONDITIONS

A. Environmental Limitations: Do not install roller shades until construction and finish work in spaces, including painting, is complete and dry and ambient temperature and humidity conditions are maintained at the levels indicated for Project when occupied for its intended use.

B. Field Measurements: Where roller shades are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Allow clearances for operating hardware of operable glazed units through entire operating range. Notify Architect of installation conditions that vary from Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

PART 2 - PRODUCTS

Manufacturers and products listed in SpecAgent and Masterworks Paragraph Builder are neither recommended nor endorsed by the AIA or ARCOM. Before inserting names, verify that manufacturers and products listed there comply with requirements retained or revised in descriptions and are both available and suitable for the intended applications. For definitions of terms and requirements for Contractor's product selection, see Section 016000 "Product Requirements."

2.1 MANUFACTURERS

A. Source Limitations: Obtain roller shades from single source from single manufacturer.

2.2 MANUALLY OPERATED SHADES WITH SINGLE ROLLERS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; Manual FlexShade or a comparable product by one of the following:

1. BTX Window Automation Inc.
2. CACO, Inc., Window Fashions.

3. DFB Sales Inc.
4. Hunter Douglas Contract.
5. Insolroll Window Shading Systems.
6. Lutron Electronics Co., Inc.
7. MechoShade Systems, Inc.
8. OEM Shades Inc.
9. Qmotion Shades.
10. Silent Gliss.
11. SM Automatic, Inc.
12. Springs Window Fashions; SWFcontract.
13. <**Insert manufacturer's name**>.

Retain "Chain-and-Clutch Operating Mechanisms" or "Crank-and-Gear Operating Mechanisms" Paragraph below or, if more than one type of manual operating mechanism is required, indicate locations of each on Drawings or in a window-treatment schedule.

B. Chain-and-Clutch Operating Mechanisms: With continuous-loop bead chain and clutch that stops shade movement when bead chain is released; permanently adjusted and lubricated.

1. Bead Chains: [**Manufacturer's standard**] [**Nickel-plated metal**] [**Stainless steel**] [**Plastic**] <**Insert description**>.

a. Loop Length: [**Full length of roller shade**] [**As indicated on Drawings**] <**Insert length**>.
b. Limit Stops: Provide upper and lower ball stops.
c. Chain-Retainer Type: [**Clip, jamb mount**] [**Chain tensioner, jamb mounted**] [**Chain tensioner, sill mounted**] <**Insert description**>.

Retain "Spring Lift-Assist Mechanisms" Subparagraph below for heavy roller shades; verify availability with manufacturers and manufacturers' weight-limit recommendations.

2. Spring Lift-Assist Mechanisms: Manufacturer's standard for balancing roller shade weight and for lifting heavy roller shades.

a. Provide for shadebands that weigh more than [**10 lb (4.5 kg)**] <**Insert value**> or for shades as recommended by manufacturer, whichever criterion is more stringent.

C. Crank-and-Gear Operating Mechanisms: Sealed gearbox drive system controlled by crank handle.

1. Crank-Handle Type: [**Detachable**] [**Permanently mounted**].
2. Crank-Handle Length: [**6 feet (1.8 m)**] [**10 feet (3 m)**] [**Manufacturer's standard**] [**As indicated on Drawings**] <**Insert length**>.

D. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

1. Roller Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with fascia, headbox, or shade-pocket design.

2. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].
3. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] [**Removable spline fitting into integral channel in tube**] <**Insert description**>.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware (for example, extended brackets or slim-profile brackets) are required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

E. Mounting Hardware: Brackets or endcaps, corrosion resistant and compatible with roller assembly, operating mechanism, installation accessories, and mounting location and conditions indicated.

F. Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

Methods of sealing light gaps at bottoms of shades vary among manufacturers.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] [**Exposed with endcaps and integral light seal at bottom where it meets the sill**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Installation Accessories:

Retain "Front Fascia" or "Exposed Headbox" Subparagraph below for exposed roller enclosures. Retain "Exposed Headbox" Subparagraph for light-blocking shades.

1. Front Fascia: Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller endcaps without exposed fasteners.

Shapes and heights of fasciae vary among manufacturers.

a. Shape: [**L-shaped**] [**Curved**] <**Insert requirements**>.

b. Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than [**4 inches (102 mm)**] [**3 inches (76 mm)**] <**Insert dimension**>.

2. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**4 inches (102 mm)**] [**3 inches (76 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.

If retaining "Endcap Covers" Subparagraph below, verify availability with manufacturers. Depending on manufacturer and manual operating system, endcaps might be required for mounting shades or attaching fasciae or headboxes.

3. Endcap Covers: To cover exposed endcaps.

Retain "Recessed Shade Pocket" Subparagraph below for roller enclosure installed above the ceiling.

4. Recessed Shade Pocket: Rectangular, extruded-aluminum enclosure designed for recessed ceiling installation; with front, top, and back formed as one piece, end plates, and removable bottom closure panel.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**6 inches (152 mm)**] [**5 inches (127 mm)**] [**4 inches (102 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.
b. Provide pocket with lip at lower edge to support acoustical ceiling panel.

5. Closure Panel and Wall Clip: Removable aluminum panel designed for installation at bottom of site-constructed ceiling recess or pocket and for snap-in attachment to wall clip without fasteners.

Widths of closure panels vary among manufacturers.

a. Closure-Panel Width: [**As indicated on Drawings**] [**2 inches (51 mm)**] <**Insert dimension**>.

Retain "Side Channels" and "Bottom (Sill) Channel or Angle" subparagraphs below for light-blocking shades.

6. Side Channels: With light seals and designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.

Methods of sealing light gaps at bottoms of shades vary among manufacturers. Shadeband bottom bars fit into bottom channels or butt against bottom angles to seal light leaks.

7. Bottom (Sill) Channel or Angle: With light seals and designed to eliminate light gaps at bottoms of shades when shades are closed.
8. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.3 MANUALLY OPERATED SHADES WITH DOUBLE ROLLERS

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; Dual Roller FlexShade or a comparable product by one of the following:

1. BTX Window Automation Inc.
2. DFB Sales Inc.
3. Hunter Douglas Contract.
4. Lutron Electronics Co., Inc.
5. MechoShade Systems, Inc.
6. OEM Shades Inc.
7. Silent Gliss.
8. SM Automatic, Inc.
9. Springs Window Fashions; SWFcontract.
10. <**Insert manufacturer's name**>.

Retain "Chain-and-Clutch Operating Mechanisms" or "Crank-and-Gear Operating Mechanisms" Paragraph below or, if more than one type of manual operating mechanism is required, indicate locations of each on Drawings or in a window-treatment schedule.

B. Chain-and-Clutch Operating Mechanisms: With continuous-loop bead chain and clutch that stops shade movement when bead chain is released; permanently adjusted and lubricated.

1. Bead Chains: [**Manufacturer's standard**] [**Nickel-plated metal**] [**Stainless steel**] [**Plastic**] <**Insert description**>.

a. Loop Length: [**Full length of roller shade**] [**As indicated on Drawings**] <**Insert length**>.
b. Limit Stops: Provide upper and lower ball stops.
c. Chain-Retainer Type: [**Clip, jamb mount**] [**Chain tensioner, jamb mounted**] [**Chain tensioner, sill mounted**] <**Insert description**>.

Retain "Spring Lift-Assist Mechanisms" Subparagraph below for heavy roller shades; verify availability with manufacturers and manufacturers' weight-limit recommendations.

2. Spring Lift-Assist Mechanisms: Manufacturer's standard for balancing roller shade weight and for lifting heavy roller shades.

a. Provide for shadebands that weigh more than [**10 lb (4.5 kg)**] <**Insert value**> or for shades as recommended by manufacturer, whichever criterion is more stringent.

C. Crank-and-Gear Operating Mechanisms: Sealed gearbox drive system controlled by crank handle.

1. Crank-Handle Type: [**Detachable**] [**Permanently mounted**].
2. Crank-Handle Length: [**6 feet (1.8 m)**] [**10 feet (3 m)**] [**Manufacturer's standard**] [**As indicated on Drawings**] <**Insert length**>.

D. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands

indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

Coordinate option retained in "Dual-Roller Mounting Configuration" Subparagraph below with available space. Side-by-side mounting is wider than offset mounting; offset mounting is taller than side-by-side mounting.

1. Dual-Roller Mounting Configuration: [**Offset, outside roller over and inside roller under**] <**Insert requirements**>.
2. Inside Roller:

    a. Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with mounting configuration, fascia, headbox, or shade-pocket design.

    b. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].

3. Outside Roller:

    a. Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with mounting configuration, fascia, headbox, or shade-pocket design.

    b. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].

4. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] [**Removable spline fitting into integral channel in tube**] <**Insert description**>.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware (for example, one-piece, dual-roller brackets) are required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

E. Mounting Hardware: Brackets or endcaps, corrosion resistant and compatible with roller mounting configuration, roller assemblies, operating mechanisms, installation accessories, and installation locations and conditions indicated.

F. Inside Shadebands:

Typically, light-filtering shadebands are installed on the inside rollers and light-blocking shadebands are installed on the outside rollers in double-roller configurations. Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Outside Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

Methods of sealing light gaps at bottoms of shades vary among manufacturers.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] [**Exposed with endcaps and integral light seal at bottom where it meets the sill**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

H. Installation Accessories:

Retain "Front Fascia" or "Exposed Headbox" Subparagraph below for exposed roller enclosures. Retain "Exposed Headbox" Subparagraph for light-blocking shades.

1. Front Fascia: Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller endcaps without exposed fasteners.

Shapes and heights of fasciae vary among manufacturers.

a. Shape: [**L-shaped**] [**Curved**] <**Insert requirements**>.
b. Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than 7 inches (178 mm).

2. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**9-1/2 inches (241 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.

If retaining "Endcap Covers" Subparagraph below, verify availability with manufacturers. Depending on manufacturer and manual operating system, endcaps might be required for mounting shades or attaching fasciae or headboxes.

3. Endcap Covers: To cover exposed endcaps.

Retain "Recessed Shade Pocket" Subparagraph below for roller enclosure installed above the ceiling.

4. Recessed Shade Pocket: Rectangular, extruded-aluminum enclosure designed for recessed ceiling installation; with front, top, and back formed as one piece, end plates, and removable bottom closure panel.

    a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**9-1/2 inches (241 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.
    b. Provide pocket with lip at lower edge to support acoustical ceiling panel.

Retain "Side Channels" and "Bottom (Sill) Channel or Angle" subparagraphs below for light-blocking shades.

5. Side Channels: With light seals and designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.

Methods of sealing light gaps at bottoms of shades vary among manufacturers. Shadeband bottom bars fit into bottom channels or butt against bottom angles to seal light leaks.

6. Bottom (Sill) Channel or Angle: With light seals and designed to eliminate light gaps at bottoms of shades when shades are closed.
7. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.4 MANUALLY OPERATED SKYLIGHT ROLLER SHADES

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; SkyLight Manual FlexShade or a comparable product by one of the following:

1. DFB Sales Inc.
2. Hunter Douglas Contract.
3. Lutron Electronics Co., Inc.
4. MechoShade Systems, Inc.
5. <**Insert manufacturer's name**>.

B. Manual Operating Mechanisms: Manufacturer's complete system and accessories suitable for conditions indicated, recommended by manufacturer for use with shade indicated, and as required for reliable operation without malfunction.

Manually operated, skylight roller shades are available from a limited number of manufacturers. Retaining "Crank-and-Gear Operating Mechanisms" Subparagraph below further limits competition among manufacturers.

1. Crank-and-Gear Operating Mechanisms: Double-roller systems with sealed gearbox drive systems controlled by detachable crank handles in drive rollers and with springs in opposing rollers to hold shadebands taught.

    a. Crank-Handle Length: [**6 feet (1.8 m)**] [**10 feet (3 m)**] [**Manufacturer's standard**] [**As indicated on Drawings**] <**Insert length**>.

C. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

1. Roller Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with fascia, headbox, or shade-pocket design.

2. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].
3. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] <**Insert description**>.

D. Shadeband Retention System: Manufacturer's standard system for guiding shadeband through range of travel and holding shadeband taut with edges of shadeband supported by side channels or angles.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware are required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

E. Mounting Hardware: Corrosion resistant and compatible with operating mechanism, installation accessories, and mounting location and conditions indicated.

F. Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Manufacturer's standard for operating mechanism indicated.

    a. Color and Finish of Exposed Bottom Bar: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Installation Accessories:

1. Exposed Headboxes and Bottom Boxes: Rectangular, extruded-aluminum enclosure including front fasciae, top and back covers, endcaps, and removable closures.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**height indicated on Drawings**] <**Insert dimension**>.

2. Channels or Angles: Manufacturer's standard design for operating mechanism indicated and shadeband take-up and support.
3. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.5 MOTOR-OPERATED, SINGLE-ROLLER SHADES

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; Motorized FlexShade or a comparable product by one of the following:

1. BTX Window Automation Inc.
2. CACO, Inc., Window Fashions.
3. DFB Sales Inc.
4. Hunter Douglas Contract.
5. Insolroll Window Shading Systems.
6. Lutron Electronics Co., Inc.
7. MechoShade Systems, Inc.
8. OEM Shades Inc.
9. Qmotion Shades.
10. Silent Gliss.
11. SM Automatic, Inc.
12. Springs Window Fashions; SWFcontract.
13. <**Insert manufacturer's name**>.

B. Motorized Operating System: Provide factory-assembled, shade-operator system of size and capacity and with features, characteristics, and accessories suitable for conditions indicated, complete with electric motor and factory-prewired motor controls, power disconnect switch, enclosures protecting controls and operating parts, and accessories required for reliable operation without malfunction. Include wiring from motor controls to motors. Coordinate operator wiring requirements and electrical characteristics with building electrical system.

1. Electrical Components: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

If required, insert manufacturers' description of alternative motor that is available (for example, "quiet operation" motor, and so forth) in "Electric Motor" Subparagraph below.

2. Electric Motor: [**Manufacturer's standard**] <**Insert description**> tubular, enclosed in roller.

a. Electrical Characteristics: [**110-V ac**] [**220-V ac**] [**24-V dc**] [**12-V dc**] [**Solar-powered dc**] <**Insert requirements**>.
b. Maximum Total Shade Width: [**As required to operate roller shades indicated**] <**Insert requirements**>.
c. Maximum Shade Drop: [**As required to operate roller shades indicated**] <**Insert requirements**>.

d. Maximum Weight Capacity: [**As required to operate roller shades indicated**] <**Insert requirements**>.

Every motor type has a different control configuration. Controls described in "Controls" subparagraph below are typical for 110v and 220v shade motors. RTS and low voltage motors use different controls and configurations. Refer to every motor type has a different control configuration.

3. Remote Control: Electric controls with NEMA ICS 6, Type 1 enclosure for [**surface**] [**recessed or flush**] <**Insert type**> mounting. Provide the following for remote-control activation of shades:

Retain one or more of first eight subparagraphs below or revise to suit Project.

a. Keyed Control Station: Keyed, [**maintained**] [**momentary**]-contact, three-position, switch-operated control station with open, close, and off functions. Provide two keys per station.
b. Individual Switch Control Station: [**Maintained**] [**Momentary**]-contact, wall-switch-operated control station with open, close, and center off functions.

1) Switch Positions: [**Three**] [**Five**].
2) Switch Style: [**Toggle**] [**Rocker**].

c. Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for single-switch group control.
d. Individual/Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for individual and group control.
e. Sun Sensor Control: Adjustable system consisting of digital displays detecting sun intensity and responding by automatically adjusting shades.
f. Infrared Control: System consisting of concealed receiver complete with external eye and connecting modular cable and [**two**] <**Insert number**> portable, multiple-channel transmitters with separate buttons to open and close individual or groups of shades, to open and close shades simultaneously, and to stop shade movement.

1) Capacity: Up to [**12**] <**Insert number**> individual or groups of shades.

g. Timer Control: Clock timer, [**24-hour**] [**seven-day**] <**Insert period**> programmable for regular events.
h. Microprocessor Control: Electronic programmable means for setting, changing, and adjusting control features; isolated from voltage spikes and surges.

Retain "Color" Subparagraph below if the color of control unit is critical.

i. Color: [**As selected by Architect from manufacturer's full range**] <**Insert color**>.

4. Limit Switches: Adjustable switches interlocked with motor controls and set to stop shades automatically at fully raised and fully lowered positions.
5. Operating Features:

a. Group switching with integrated switch control; single faceplate for multiple switch cutouts.

b. Capable of interface with [**audiovisual**] [**multiroom**] <**Insert description**> control system.
c. Capable of accepting input from building automation control system.

Retain first subparagraph below only for sensor-, infrared-, or timer-controlled systems.

d. Override switch.

6. Accessories:

a. <**Insert accessory**>.

C. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

1. Roller Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with fascia, headbox, or shade-pocket design.

2. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].
3. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] [**Removable spline fitting into integral channel in tube**] <**Insert description**>.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware (for example, extended brackets or slim-profile brackets) are required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

D. Mounting Hardware: Brackets or endcaps, corrosion resistant and compatible with roller assembly, operating mechanism, installation accessories, and mounting location and conditions indicated.

If retaining "Roller-Coupling Assemblies" Paragraph below, indicate locations and quantities of rollers joined with coupling assemblies on Drawings or in a window-treatment schedule.

E. Roller-Coupling Assemblies: Coordinated with operating mechanism and designed to join up to three inline rollers that are operated by one roller drive-end assembly.

F. Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

Methods of sealing light gaps at bottoms of shades vary among manufacturers. Exposed bottom bars with integral light seals are available from MechoShade.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] [**Exposed with endcaps and integral light seal at bottom where it meets the sill**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Installation Accessories:

Retain "Front Fascia" or "Exposed Headbox" Subparagraph below for exposed roller enclosures. Retain "Exposed Headbox" Subparagraph for light-blocking shades.

1. Front Fascia: Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller endcaps without exposed fasteners.

Shapes and heights of fasciae vary among manufacturers.

a. Shape: [**L-shaped**] [**Curved**] <**Insert requirements**>.
b. Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than [**4 inches (102 mm)**] [**3 inches (76 mm)**] <**Insert dimension**>.

2. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure.

a. Height: Manufacturer's standard in height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**4 inches (102 mm)**] [**3 inches (76 mm)**] [**as indicated on Drawings**] <**Insert dimension**>.

If retaining "Endcap Covers" Subparagraph below, verify availability with manufacturers.

3. Endcap Covers: To cover exposed endcaps.

Retain "Recessed Shade Pocket" Subparagraph below for roller enclosure installed above the ceiling.

4. Recessed Shade Pocket: Rectangular, extruded-aluminum enclosure designed for recessed ceiling installation; with front, top, and back formed as one piece, end plates, and removable bottom closure panel.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**6 inches (152 mm)**] [**5 inches (127 mm)**] [**4 inches (102 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.
b. Provide pocket with lip at lower edge to support acoustical ceiling panel.

5. Closure Panel and Wall Clip: Removable aluminum panel designed for installation at bottom of site-constructed ceiling recess or pocket and for snap-in attachment to wall clip without fasteners.

Widths of closure panels vary among manufacturers.

a. Closure-Panel Width: [**As indicated on Drawings**] [**2 inches (51 mm)**] <**Insert dimension**>.

Retain "Side Channels" and "Bottom (Sill) Channel or Angle" subparagraphs below for light-blocking shades.

6. Side Channels: With light seals and designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.

Methods of sealing light gaps at bottoms of shades vary among manufacturers. Shadeband bottom bars fit into bottom channels or butt against bottom angles to seal light leaks.

7. Bottom (Sill) Channel or Angle: With light seals and designed to eliminate light gaps at bottoms of shades when shades are closed.
8. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.6 MOTOR-OPERATED, DUAL-ROLLER SHADES

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; Dual Roller FlexShade or a comparable product by one of the following:

1. BTX Window Automation Inc.
2. DFB Sales Inc.
3. Hunter Douglas Contract.
4. Insolroll Window Shading Systems.
5. Lutron Electronics Co., Inc.
6. MechoShade Systems, Inc.
7. OEM Shades Inc.
8. Silent Gliss.
9. SM Automatic, Inc.
10. Springs Window Fashions; SWFcontract.
11. <**Insert manufacturer's name**>.

B. Motorized Operating Systems: Provide factory-assembled, shade-operator systems of size and capacity and with features, characteristics, and accessories suitable for conditions indicated, complete with electric motor and factory-prewired motor controls, power disconnect switch, enclosures protecting controls and operating parts, and accessories required for reliable operation without malfunction. Include wiring from motor controls to motors. Coordinate operator wiring requirements and electrical characteristics with building electrical system.

1. Electrical Components: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

If required, insert manufacturers' description of alternative motor that is available (for example, "quiet operation" motor, and so forth) in "Electric Motor" Subparagraph below.

2. Electric Motor: [**Manufacturer's standard**] <**Insert description**> tubular, enclosed in rollers.

a. Electrical Characteristics: [**110-V ac**] [**220-V ac**] [**24-V dc**] [**12-V dc**] [**Solar-powered dc**] <**Insert requirements**>.
b. Maximum Total Shade Width: [**As required to operate roller shades indicated**] <**Insert requirements**>.
c. Maximum Shade Drop: [**As required to operate roller shades indicated**] <**Insert requirements**>.
d. Maximum Weight Capacity: [**As required to operate roller shades indicated**] <**Insert requirements**>.

Every motor type has a different control configuration. Controls described in "Controls" subparagraph below are typical for 110v and 220v shade motors. RTS and low voltage motors use different controls and configurations. Refer to every motor type has a different control configuration.

3. Remote Control: Electric controls with NEMA ICS 6, Type 1 enclosure for [**surface**] [**recessed or flush**] <**Insert type**> mounting. Provide the following for remote-control activation of shades:

Retain one or more of first eight subparagraphs below or revise to suit Project.

a. Keyed Control Station: Keyed, [**maintained**] [**momentary**]-contact, three-position, switch-operated control station with open, close, and off functions. Provide two keys per station.
b. Individual Switch Control Station: [**Maintained**] [**Momentary**]-contact, wall-switch-operated control station with open, close, and center off functions.

1) Switch Positions: [**Three**] [**Five**].
2) Switch Style: [**Toggle**] [**Rocker**].

c. Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for single-switch group control.
d. Individual/Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for individual and group control.
e. Sun Sensor Control: Adjustable system consisting of digital displays detecting sun intensity and responding by automatically adjusting shades.
f. Infrared Control: System consisting of concealed receiver complete with external eye and connecting modular cable and [**two**] <**Insert number**> portable, multiple-channel transmitters with separate buttons to open and close individual or groups of shades, to open and close shades simultaneously, and to stop shade movement.

1) Capacity: Up to [**12**] <**Insert number**> individual or groups of shades.

g. Timer Control: Clock timer, [**24-hour**] [**seven-day**] <**Insert period**> programmable for regular events.
h. Microprocessor Control: Electronic programmable means for setting, changing, and adjusting control features; isolated from voltage spikes and surges.

Retain "Color" Subparagraph below if the color of control unit is critical.

i. Color: [**As selected by Architect from manufacturer's full range**] <**Insert color**>.

4. Limit Switches: Adjustable switches, interlocked with motor controls and set to stop shade movement automatically at fully raised and fully lowered positions.
5. Operating Features:

    a. Group switching with integrated switch control; single faceplate for multiple switch cutouts.
    b. Capable of interface with [**audiovisual**] [**multiroom**] <**Insert description**> control system.
    c. Capable of accepting input from building automation control system.

Retain first subparagraph below only for sensor-, infrared-, or timer-controlled systems.

    d. Override switch.

6. Accessories:

    a. <**Insert accessory**>.

C. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shades for service.

Coordinate option retained in "Dual-Roller Mounting Configuration" Subparagraph below with available space.

1. Dual-Roller Mounting Configuration: [**Offset, outside shade over and inside shade under**] <**Insert requirements**>.
2. Inside Roller:

    a. Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with mounting configuration, fascia, headbox, or shade-pocket design.

    b. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].

3. Outside Roller:

    a. Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with mounting configuration, fascia, headbox, or shade-pocket design.

    b. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].

4. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] [**Removable spline fitting into integral channel in tube**] <**Insert description**>.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware (for example, one-piece, double-roller brackets) are

required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

D. Mounting Hardware: Brackets or endcaps, corrosion resistant and compatible with roller mounting configuration, roller assemblies, operating mechanisms, installation accessories, and installation locations and conditions indicated.

If retaining "Roller-Coupling Assemblies" Paragraph below, indicate locations and quantities of rollers joined with coupling assemblies on Drawings or in a window-treatment schedule.

E. Roller-Coupling Assemblies: Coordinated with operating mechanism and designed to join up to three inline rollers into a multiband shade that is operated by one roller drive-end assembly.

F. Inside Shadebands:

Typically, light-filtering shadebands are installed on the inside rollers and light-blocking shadebands are installed on the outside rollers in double-roller configurations. Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Outside Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Steel or extruded aluminum.

Bottom bars vary among manufacturers; insert requirements for specific type and shape in "Type" Subparagraph below.

a. Type: [**Enclosed in sealed pocket of shadeband material**] [**Exposed with endcaps**] [**Exposed with endcaps and integral light seal where bottom (sill) channels are indicated**] <**Insert description**>.

Retain "Color and Finish" Subparagraph below for exposed bottom bars.

b. Color and Finish: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

H. Installation Accessories:

Retain "Front Fascia" or "Exposed Headbox" Subparagraph below for exposed roller enclosures. Retain "Exposed Headbox" Subparagraph for light-blocking shades.

1. Front Fascia: Aluminum extrusion that conceals front and underside of roller and operating mechanism and attaches to roller endcaps without exposed fasteners.

Shapes and heights of fasciae vary among manufacturers.

a. Shape: [**L-shaped**] [**Curved**] <**Insert requirements**>.
b. Height: Manufacturer's standard height required to conceal roller and shadeband assembly when shade is fully open, but not less than 7 inches (178 mm).

2. Exposed Headbox: Rectangular, extruded-aluminum enclosure including front fascia, top and back covers, endcaps, and removable bottom closure.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**9-1/2 inches (241 mm)**] [**3 inches (76 mm)**] [**as indicated on Drawings**] <**Insert dimension**>.

If retaining "Endcap Covers" Subparagraph below, verify availability with manufacturers.

3. Endcap Covers: To cover exposed endcaps.

Retain "Recessed Shade Pocket" Subparagraph below for roller enclosure installed above the ceiling.

4. Recessed Shade Pocket: Rectangular, extruded-aluminum enclosure designed for recessed ceiling installation; with front, top, and back formed as one piece, end plates, and removable bottom closure panel.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**6 inches (152 mm)**] [**5 inches (127 mm)**] [**4 inches (102 mm)**] [**height indicated on Drawings**] <**Insert dimension**>.
b. Provide pocket with lip at lower edge to support acoustical ceiling panel.

5. Closure Panel and Wall Clip: Removable aluminum panel designed for installation at bottom of site-constructed ceiling recesses or pockets and for snap-in attachment to wall clip without fasteners.

Widths of closure panels vary among manufacturers.

a. Closure-Panel Width: [**As indicated on Drawings**] [**2 inches (51 mm)**] <**Insert dimension**>.

Retain "Side Channels" and "Bottom (Sill) Channel or Angle" subparagraphs below for light-blocking shades.

6. Side Channels: With light seals and designed to eliminate light gaps at sides of shades as shades are drawn down. Provide side channels with shadeband guides or other means of aligning shadebands with channels at tops.

Methods of sealing light gaps at bottoms of shades vary among manufacturers. Shadeband bottom bars fit into bottom channels or butt against bottom angles to seal light leaks.

7. Bottom (Sill) Channel or Angle: With light seals and designed to eliminate light gaps at bottoms of shades when shades are closed.
8. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.7 MOTOR-OPERATED, SKYLIGHT ROLLER SHADES

A. Basis-of-Design Product: Subject to compliance with requirements, provide Draper Inc.; SkyLight 2 FlexShade or a comparable product by one of the following:

1. DFB Sales Inc.
2. Hunter Douglas Contract.
3. Lutron Electronics Co., Inc.
4. MechoShade Systems, Inc.
5. <**Insert manufacturer's name**>.

B. Motorized Operating System: Provide factory-assembled, shade-operator system of size and capacity and with features, characteristics, and accessories suitable for conditions indicated, complete with electric motor and factory-prewired motor controls, power disconnect switch, enclosures protecting controls and operating parts, and accessories required for reliable operation without malfunction. Include wiring from motor controls to motors. Coordinate operator wiring requirements and electrical characteristics with building electrical system.

1. System Type: [**Single-motor, double-roller system, with motor in drive roller and spring in take-up roller**] [**Single motor and roller, gravity drop**] <**Insert description**>.
2. Electrical Components: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

If required, insert manufacturers' description of alternative motor that is available (for example, "quiet operation" motor, and so forth) in "Electric Motor" Subparagraph below.

3. Electric Motor: [**Manufacturer's standard**] <**Insert description**> tubular, enclosed in roller.

    a. Electrical Characteristics: [**110-V ac**] [**220-V ac**] [**24-V dc**] [**12-V dc**] [**Solar-powered dc**] <**Insert requirements**>.
    b. Maximum Total Shade Width: [**As required to operate roller shades indicated**] <**Insert requirements**>.
    c. Maximum Shade Length: [**As required to operate roller shades indicated**] <**Insert requirements**>.
    d. Maximum Weight Capacity: [**As required to operate roller shades indicated**] <**Insert requirements**>.

Every motor type has a different control configuration. Controls described in "Controls" subparagraph below are typical for 110v and 220v shade motors. RTS and low voltage motors use different controls and configurations. Refer to every motor type has a different control configuration.

4. Remote Control: Electric controls with NEMA ICS 6, Type 1 enclosure for [**surface**] [**recessed or flush**] <**Insert type**> mounting. Provide the following for remote-control activation of shades:

Retain one or more of first eight subparagraphs below or revise to suit Project.

a. Keyed Control Station: Keyed, [**maintained**] [**momentary**]-contact, three-position, switch-operated control station with open, close, and off functions. Provide two keys per station.
b. Individual Switch Control Station: [**Maintained**] [**Momentary**]-contact, wall-switch-operated control station with open, close, and center off functions.

1) Switch Positions: [**Three**] [**Five**].
2) Switch Style: [**Toggle**] [**Rocker**].

c. Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for single-switch group control.
d. Individual/Group Control Station: [**Maintained**] [**Momentary**]-contact, three-position, rocker-style, wall-switch-operated control station with open, close, and center off functions for individual and group control.
e. Sun Sensor Control: Adjustable system consisting of digital displays detecting sun intensity and responding by automatically adjusting shades.
f. Infrared Control: System consisting of concealed receiver complete with external eye and connecting modular cable and [**two**] <**Insert number**> portable, multiple-channel transmitters with separate buttons to open and close individual or groups of shades, to open and close shades simultaneously, and to stop shade movement.

1) Capacity: Up to [**12**] <**Insert number**> individual or groups of shades.

g. Timer Control: Clock timer, [**24-hour**] [**seven-day**] <**Insert period**> programmable for regular events.
h. Microprocessor Control: Electronic programmable means for setting, changing, and adjusting control features; isolated from voltage spikes and surges.

Retain "Color" Subparagraph below if the color of the control unit is critical.

i. Color: [**As selected by Architect from manufacturer's full range**] <**Insert color**>.

5. Limit Switches: Adjustable switches, interlocked with motor controls and set to stop shade movement automatically at fully raised and fully lowered positions.
6. Operating Features:

a. Group switching with integrated switch control; single faceplate for multiple switch cutouts.
b. Capable of interface with [**audiovisual**] [**multiroom**] <**Insert description**> control system.
c. Capable of accepting input from building automation control system.

Retain first subparagraph below only for sensor-, infrared-, or timer-controlled systems.

d. Override switch.

7. Accessories:

a. <**Insert accessory**>.

C. Rollers: Corrosion-resistant steel or extruded-aluminum tubes of diameters and wall thicknesses required to accommodate operating mechanisms and weights and widths of shadebands indicated without deflection. Provide with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

1. Drive-End Location: [**Right side of interior face of shade**] [**Left side of interior face of shade**] [**As indicated on Drawings**] <**Insert requirements**>.

Coordinate direction of roll with fascia, headbox, or shade-pocket design.

2. Direction of Shadeband Roll: [**Regular, from back (exterior face) of roller**] [**Reverse, from front (interior face) of roller**].
3. Shadeband-to-Roller Attachment: [**Manufacturer's standard method**] [**Adhesive strip**] <**Insert description**>.

D. Shadeband Retention System: Manufacturer's standard system for guiding shadeband through range of travel and holding shadeband taut with edges of shadeband supported by side channels or angles.

Types and sizes of mounting hardware vary among manufacturers. Revise "Mounting Hardware" Paragraph below if specific types of hardware are required. Where hardware dimensions are critical, indicate installation conditions and size constraints on Drawings.

E. Mounting Hardware: Corrosion resistant and compatible with operating mechanism, installation accessories, and mounting location and conditions indicated.

F. Shadebands:

Coordinate option retained in "Shadeband Material" Subparagraph below with requirements specified in "Shadeband Materials" Article.

1. Shadeband Material: [**Light-filtering fabric**] [**Light-blocking fabric**] <**Insert requirements**>.
2. Shadeband Bottom (Hem) Bar: Manufacturer's standard for operating mechanism indicated.

a. Color and Finish of Exposed Bottom Bar: [**As selected by Architect from manufacturer's full range**] <**Insert color and finish**>.

G. Installation Accessories:

1. Exposed Headboxes and Bottom Boxes: Rectangular, extruded-aluminum enclosure including front fasciae, top and back covers, endcaps, and removable closures.

a. Height: Manufacturer's standard height required to enclose roller and shadeband assembly when shade is fully open, but not less than [**height indicated on Drawings**] <**Insert dimension**>.

2. Channels or Angles: Manufacturer's standard design for operating mechanism indicated and shadeband take-up and support.
3. Installation Accessories Color and Finish: [**As selected from manufacturer's full range**] <**Insert color and finish**>.

2.8 SHADEBAND MATERIALS

A. Shadeband Material Flame-Resistance Rating: Comply with [**NFPA 701**] <**Insert requirement**>. Testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

If more than one light-filtering or light-blocking shadeband fabric is required, copy and re-edit applicable paragraph below and insert a unique designation for each fabric.

B. Light-Filtering Fabric: Woven fabric, stain and fade resistant.

1. Source: [**Roller shade manufacturer**] <**Insert source for custom fabrics**>.
2. Type: [**PVC-coated fiberglass**] [**PVC-coated polyester**] [**Woven PVC-coated fiberglass and PVC-coated polyester**] [**Woven polyester and PVC-coated polyester**] [**Acrylic-coated fiberglass**] [**PVC-coated fiberglass with silver backing**] <**Insert description**>.
3. Weave: [**Mesh**] [**Basketweave**] <**Insert description**>.
4. Thickness: <**Insert thickness**>.
5. Weight: <**Insert oz./sq. yd. (g/sq. m)**>.

Coordinate requirements retained in "Orientation on Shadeband" Subparagraph below with requirements in "Roller Shade Fabrication" Article. See "Shadebands" Article in the Evaluations for a discussion of up-the-bolt and railroaded fabrics.

6. Orientation on Shadeband: [**Up the bolt**] [**Railroaded**] [**As indicated on Drawings**] <**Insert requirements**>.
7. Openness Factor: [**1**] [**3**] [**5**] [**10**] [**11**] [**22**] <**Insert number**> percent.
8. Color: [**As indicated on Drawings**] [**Match Architect's sample**] [**As selected by Architect from manufacturer's full range**] <**Insert color**>.

C. Light-Blocking Fabric: Opaque fabric, stain and fade resistant.

1. Source: [**Roller shade manufacturer**] <**Insert source for custom fabrics**>.
2. Type: [**Fiberglass textile with PVC film bonded to both sides**] [**Fiberglass with acrylic backing**] [**Acrylic-coated fiberglass**] [**Polyester-cotton blend**] [**Polyester with foamed-acrylic backing**] [**PVC-coated fiberglass with bonded PVC film**] <**Insert description**>.
3. Thickness: <**Insert thickness**>.
4. Weight: <**Insert oz./sq. yd. (g/sq. m)**>.

Coordinate requirements retained in "Orientation on Shadeband" Subparagraph below with requirements in "Roller Shade Fabrication" Article. See "Shadebands" Article in the Evaluations for a discussion of up-the-bolt and railroaded fabrics.

5. Orientation on Shadeband: [**Up the bolt**] [**Railroaded**] [**As indicated on Drawings**] <**Insert requirements**>.
6. Features: [**Washable**] [**Antistatic treatment**] <**Insert requirements**>.

Outside face of light-blocking fabrics are often white.

7. Color: [**As indicated on Drawings**] [**Match Architect's sample**] [**As selected by Architect from manufacturer's full range**] <**Insert color**>.

If required, insert requirements for dual-fabric shadebands made of two layers of different, loosely layered material.

2.9 ROLLER SHADE FABRICATION

A. Product Safety Standard: Fabricate roller shades to comply with WCMA A 100.1, including requirements for flexible, chain-loop devices; lead content of components; and warning labels.

B. Unit Sizes: Fabricate units in sizes to fill window and other openings as follows, measured at 74 deg F (23 deg C):

Retain "Between (Inside) Jamb Installation" or "Outside of Jamb Installation" Subparagraph below. Alternatively, retain both subparagraphs and indicate locations of each on Drawings or in a window-treatment schedule. Coordinate clearance requirements with distance between shades and glass, glass type, and placement of heating/cooling air supplies to avoid heat buildup and possible damage to glass.

If retaining "Between (Inside) Jamb Installation" Subparagraph, see "Product Characteristics" Article in the Evaluations for a discussion of GANA recommendations.

1. Between (Inside) Jamb Installation: Width equal to jamb-to-jamb dimension of opening in which shade is installed less 1/4 inch (6 mm) per side or 1/2-inch (13-mm) total, plus or minus 1/8 inch (3.1 mm). Length equal to head-to-sill or -floor dimension of opening in which shade is installed less 1/4 inch (6 mm), plus or minus 1/8 inch (3.1 mm).

If retaining "Outside of Jamb Installation" Subparagraph below, indicate locations of shades relative to openings and widths and lengths of shades on Drawings or in a window-treatment schedule.

2. Outside of Jamb Installation: Width and length as indicated, with terminations between shades of end-to-end installations at centerlines of mullion or other defined vertical separations between openings.

C. Shadeband Fabrication: Fabricate shadebands without battens or seams to extent possible, except as follows:

1. Vertical Shades: Where width-to-length ratio of shadeband is equal to or greater than [**1:4**] <**Insert ratio**>, provide battens and seams at uniform spacings along shadeband length to ensure shadeband tracking and alignment through its full range of movement without distortion of the material.
2. Skylight Shades: Provide battens and seams at uniform spacings along shadeband as required to ensure shadeband tracking and alignment through its full range of movement without distortion or sag of material.

Coordinate requirements in "Railroaded Materials" Subparagraph below with requirements in "Shadeband Materials" Article and with requirements indicated on Drawings or in a window-treatment schedule.

3. Railroaded Materials: Railroad material where material roll width is less than the required width of shadeband and where indicated. Provide battens and seams as required by

railroaded material to produce shadebands with full roll-width panel(s) plus, if required, one partial roll-width panel located at top of shadeband.

PART 3 - EXECUTION

3.1 EXAMINATION

Retain option in first paragraph below for motorized operators.

A. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, operational clearances, [**locations of connections to building electrical system,**]and other conditions affecting performance of the Work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 ROLLER SHADE INSTALLATION

A. Install roller shades level, plumb, and aligned with adjacent units according to manufacturer's written instructions.

Coordinate requirements for distance between roller shades and glass with glass type and placement of heating/cooling air supplies to avoid heat buildup and possible damage to glass. Generally, retain first option in "Opaque Shadebands" Subparagraph below. See "Product Characteristics" Article in the Evaluations.

1. Opaque Shadebands: Located so shadeband is not closer than [**2 inches (51 mm)**] <**Insert dimension**> to interior face of glass. Allow clearances for window operation hardware.

Retain "Electrical Connections" Paragraph below for motor-operated roller shades.

B. Electrical Connections: Connect motor-operated roller shades to building electrical system.

C. Roller Shade Locations: [**At exterior windows**] [**As indicated in window-covering schedule**] [**As indicated on Drawings**] <**Insert requirements**>.

3.3 ADJUSTING

A. Adjust and balance roller shades to operate smoothly, easily, safely, and free from binding or malfunction throughout entire operational range.

3.4 CLEANING AND PROTECTION

A. Clean roller shade surfaces, after installation, according to manufacturer's written instructions.

B. Provide final protection and maintain conditions, in a manner acceptable to manufacturer and Installer, that ensure that roller shades are without damage or deterioration at time of Substantial Completion.

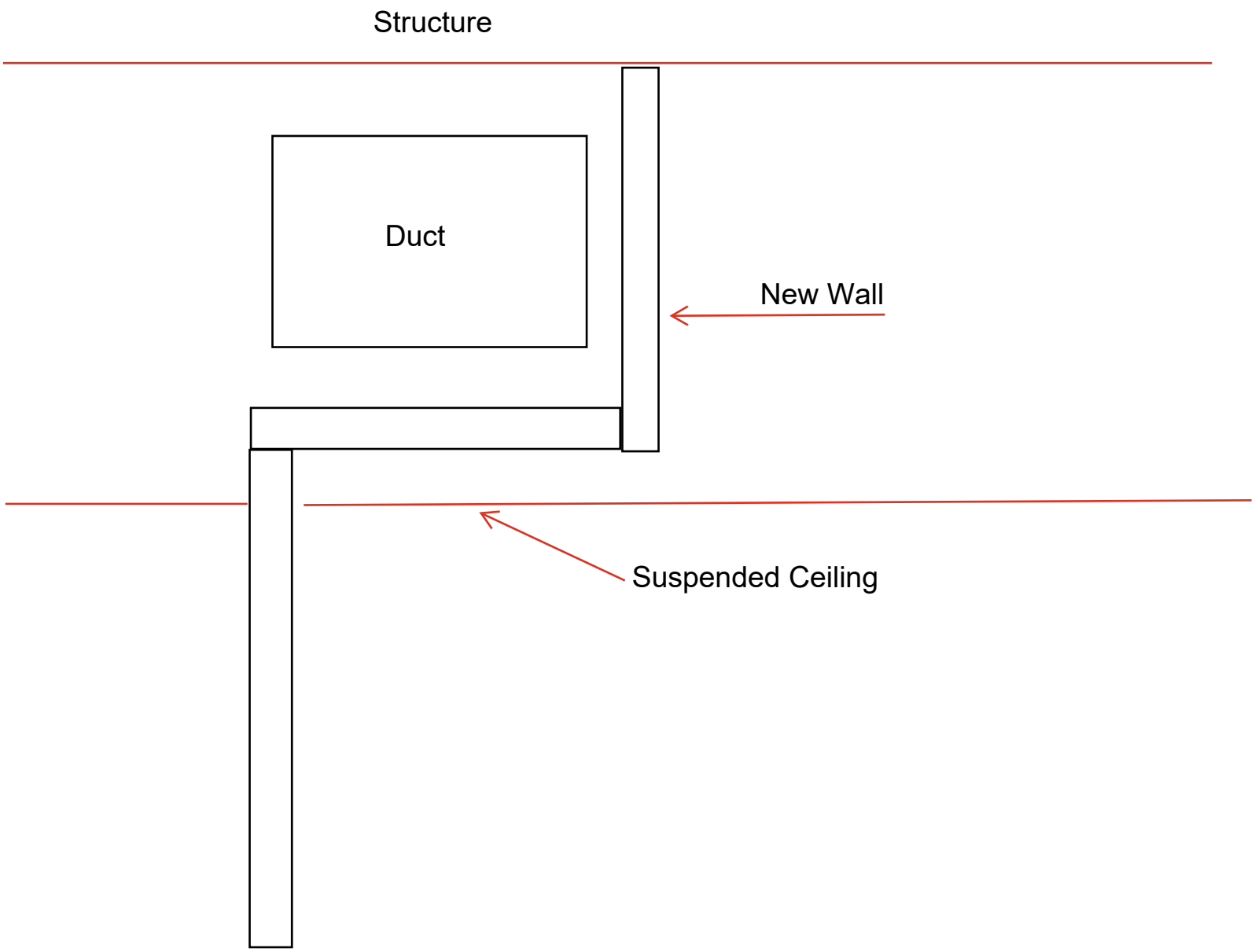
C. Replace damaged roller shades that cannot be repaired, in a manner approved by Architect, before time of Substantial Completion.

3.5 DEMONSTRATION

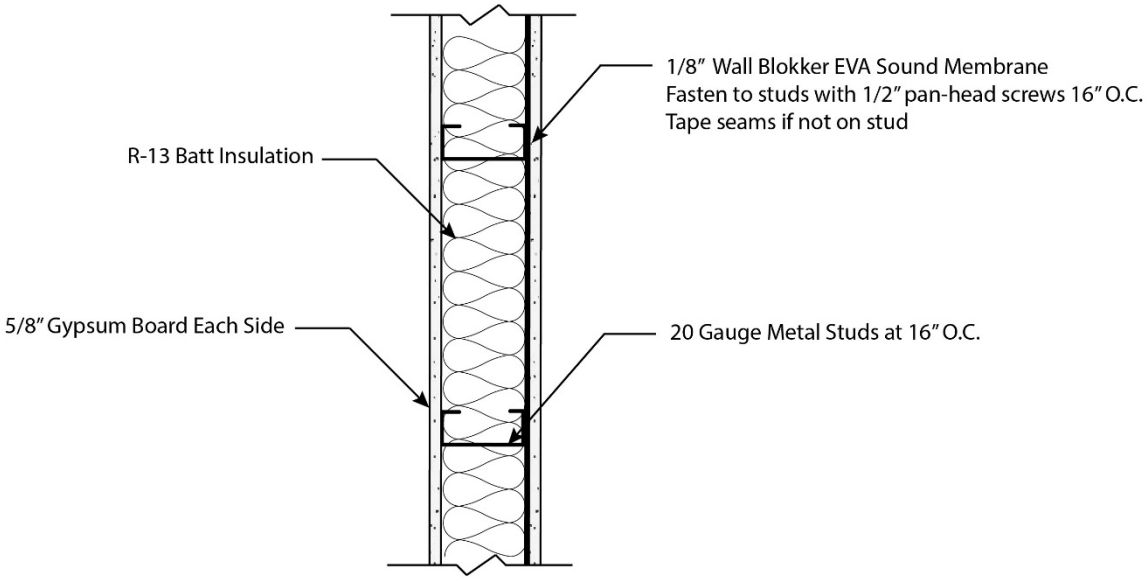
Retain this article for motorized roller shades.

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain motor-operated roller shades.

END OF SECTION 122413


Structure
Duct
New Wall
Suspended Ceiling

Partition Type CA 4



Partition	Stud Size	Fire Rating	UL	STC Rating	STC Test	Partition Thickness
CA 4-1	3-5/8"	1 hr	UL U419	STC 49	INSUL v8.0.12	4-15/16"
CA 4-2	6"	1hr	UL U419	STC 52	INSUL v8.0.12	7-3/8"

Sound Insulation Prediction (v8.0.12)

Program copyright Marshall Day Acoustics 2015

- Key No. 4851

Margin of error is generally within STC +/- 3 dB

Job Name:

Notes:

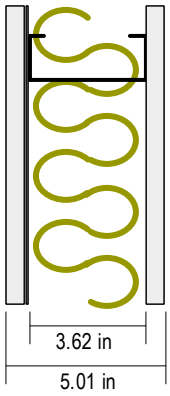
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Date: 10 Oct 17

Initials:Walker

File Name: insul



STC 49

OITC 29

System description

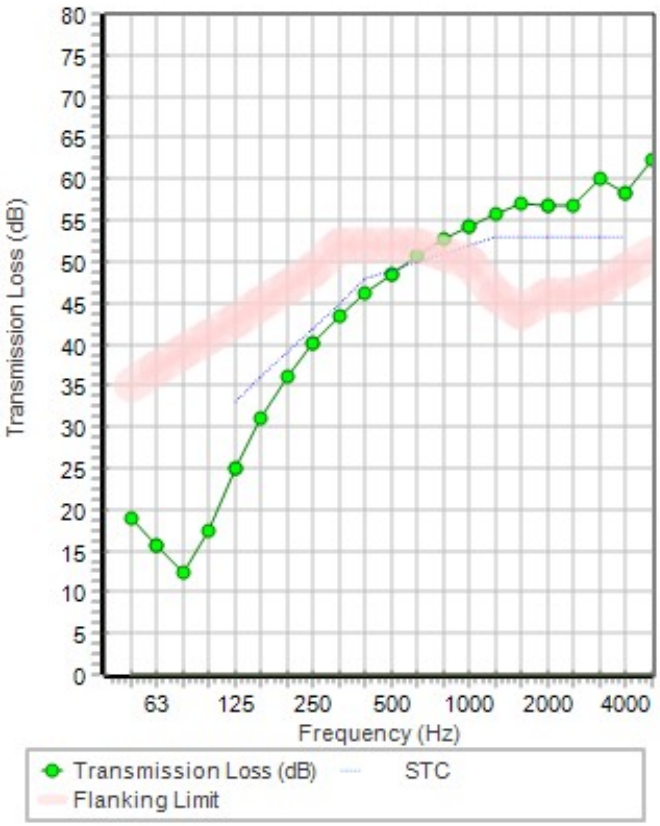
Panel 1 : 1 x 0.63 in Type X Gypsum Board + 1 x 0.13 in Wall Blokker

Cavity : Steel stud (20g-16g): Stud spacing 16 in , Infill Fibreglass (10kg/m3) Thickness 3.5 in (ρ:10 lbs/ft3, Rf:4000 Pa.s/m2)
Panel 2 + 1 x 0.63 in Type X Gypsum Board (ρ:43.08 lbs/ft3,E:0.27psi*10^6,η:0.01, ps:2.26 lb/ft2, fc:2491 Hz)

Mass-air-mass resonant frequency =66 Hz

frequency (Hz)	TL(dB)	TL(dB)
50	19	
63	16	15
80	12	
100	17	
125	25	21
160	31	
200	36	
250	40	39
315	43	
400	46	
500	48	48
630	51	
800	53	
1000	54	54
1250	56	
1600	57	
2000	57	57
2500	57	
3150	60	
4000	58	60
5000	62	

Panel Size 8.9x13 ft; Mass 6.4 lb/ft2



Sound Insulation Prediction (v8.0.12)

Program copyright Marshall Day Acoustics 2015

- Key No. 4851

Margin of error is generally within STC +/- 3 dB

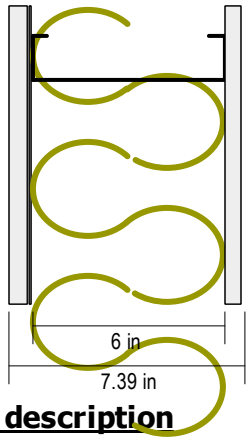
Job Name:

Notes:

Job No.: Page No.:

Date: 2 Mar 18 Initials:Walker

File Name: insul



STC 52

OITC 32

System description

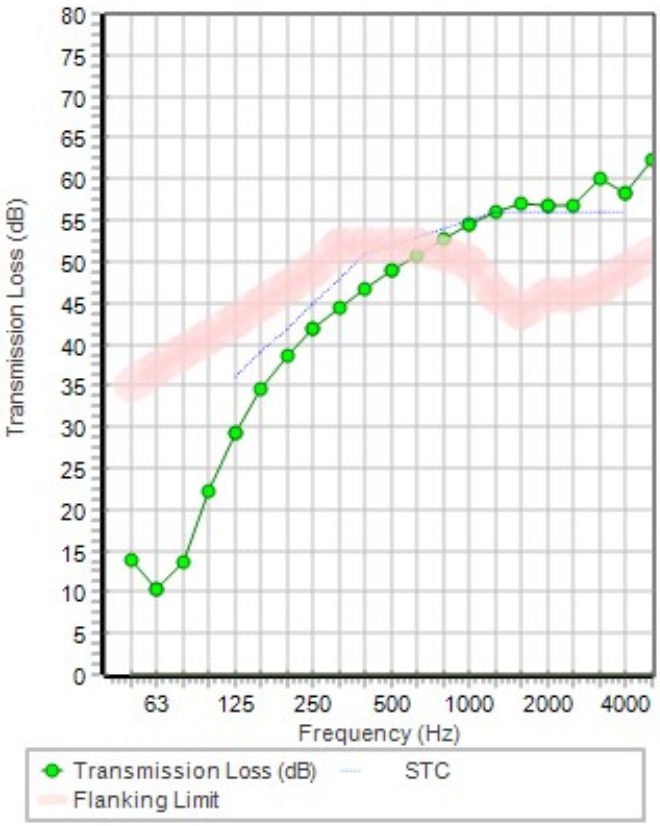
Panel 1 : 1 x 0.63 in Type X Gypsum Board + 1 x 0.13 in Wall Blokker

Cavity: Steel stud (20g-16g): Stud spacing 16 in , Infill Fibreglass (10kg/m3) Thickness 6.0 in (ρ:10 lbs/ft3, Rf:4000 Pa.s/m2)
Panel 2 + 1 x 0.63 in Type X Gypsum Board (ρ:43.08 lbs/ft3,E:0.27psi*10^6,η:0.01, ρs:2.26 lb/ft2, fc:2491 Hz)

Mass-air-mass resonant frequency =51 Hz

frequency (Hz)	TL(dB)	TL(dB)
50	14	
63	10	12
80	14	
100	22	
125	29	26
160	35	
200	39	
250	42	41
315	44	
400	47	
500	49	49
630	51	
800	53	
1000	54	54
1250	56	
1600	57	
2000	57	57
2500	57	
3150	60	
4000	58	60
5000	62	

Panel Size 8.9x13 ft; Mass 6.5 lb/ft2



Technical Data

Wall Blokker™

Acoustical Sound Barrier

World's First Soundproofing Membrane designed specifically to increase sound attenuation across walls – Hit Higher STCs at a fraction of the Cost

Commercial Acoustics Wall Blokker is an engineered barrier for sound reduction. Wall Blokker is an EVA mass loaded barrier, which is flexible and designed for commercial, industrial and residential applications. It is used behind a finished wall to reduce noise transfer across a wide array of frequencies, improving the STC of the partition.

Unlike traditional Mass Loaded Vinyl, Wall Blokker was developed in the automotive industry to provide superior soundproofing performance with long life-cycles and without off-gassing and other degrading processes. At 1/8" thick, a single layer can provide equivalent soundproofing to multiple layers of drywall, often at a fraction of the price and schedule.



Product Specifications:

- Made with an engineered recycled Acoustical Grade Polymer.
- EVA (Ethylene Vinyl Acetate) Base - does not require plasticizers
- Non PVC (no ozone depleting gasses) and no VOCs
- 3-in-1 barrier (sound, moisture and air)
- Mold and mildew resistant
- Delta-STC of 8-12 points, depending on Wall Type
- High STC performance in single and multiple layer applications
- Flexible & easy to cut and install
- Made in the USA

Product Performance:

- Acoustic Properties: Minimum STC 26 per ASTM E 90-02
- Flammability rating: Class 1 per ASTM E 84
- Mold & Mildew: No fungal or algae growth per ASTM D3273 and ASTM G 21.
- Fire Resistance: Rated for 1hr and 2 hr walls per ASTM E 119-08.
- Standard Sheet Dimensions
  - 4' by 8' sheet size
  - 4' by 25' roll size
  - Factory Cut to Wall Length to Minimize Scrap
- Sheet Weight
  - 1/8" Thick
  - 1 lb/ft^2 nominal
- Standard Tolerances
  - Width: + 0.5" - 0"
  - Length: +1% - 0"
  - Nominal Thickness: ±0.10"

Tested and approved for use in all wall designs of the U300, U400, and V400 series.

COMMERCIAL ACOUSTICS
6122 Benjamin Rd. · Tampa, Florida 33634 · 888-815-9691
www.Commercial-Acoustics.com · info@commercial-acoustics.com

Acoustic Test Data:

Full test report library and specification packets available on request

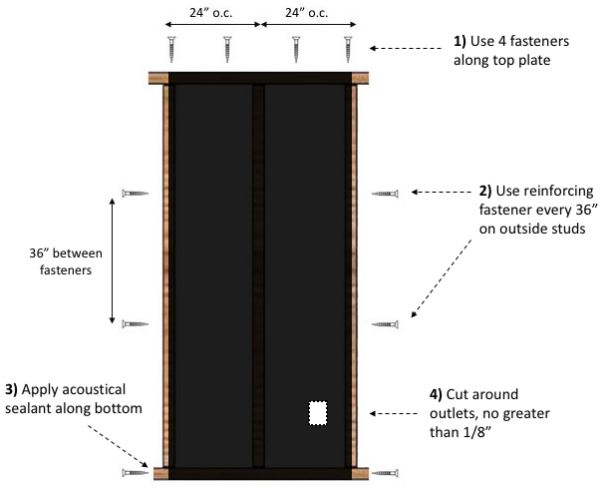
Frequency (Hz)	125	250	500	1000	2000	4000	STC	Test Number
Baseline Wood Studs	21	31	39	51	38	50	**34**	IRC-IR-693
Wall Blokker 1 side, 16" oc	17	35	42	50	47	57	**41**	RAL-TL08-138
Baseline Metal Studs	22	42	56	62	48	50	**44**	RAL-TL08-183
Wall Blokker 1 side, 16" oc	28	42	55	60	58	61	**53**	RAL-TL08-184
Wall Blokker 1 side, 24" oc	34	45	56	60	57	62	**56**	RAL-TL08-152
Wall Blokker 2 sides, 24" oc	42	53	60	64	64	70	**62**	RAL-TL08-155

Installation:

1. Ensure surface is clean and dry
2. Wall Blokker attaches to wood studs with staples or screws, and metal studs with pan-head screws
3. Roll the Wall Blokker to the proper length of the wall, measured vertically
4. Start at the top of the wall, securing the Wall Blokker to the header stud with 4-5 staples or screws
5. Roll the membrane downward, so that each edge is directly in the middle of the adjacent stud. Do not lap edges.
6. Secure the membrane to the perimeter studs at 36" inches, nominally
7. Taping of the edges is optional
    a. Mandatory if the edge does not fall on the stud face
8. Use standard drywall screws when fastening drywall board to the studs through the membrane. Wall Blokker is an EVA polymer that will tighten around the screws like a gasket
9. If holes/tears should take place, simply place vinyl tape over the gap
10. At bottom and top of the wall, ensure that gaps are less than or equal to ¼". Fill in gaps with non-hardening caulk
11. For HVAC, plumbing, or electrical penetrations, fill gap with fiberglass batting as needed and close with caulk
12. For overhead placement, placement on top of drywall prior to installation is recommended
13. Installation should not begin until all other trades are finished in the area
14. It is recommended that areas to receive Wall Blokker be weather tight. Materials can be stiff and less pliable at low temperatures

Installation Overview:

- Installs vertically, directly to studs, beneath the drywall
- Installation requires two or more capable technicians/hangers
- No special training required





Tested and approved for use in all wall designs of the U300, U400, and V400 series.

Commercial Acoustics Wall Blokker Specification

Division 09 – Finishes
Section 098400 – Acoustical Components

PART 1 – GENERAL

1.1 SUMMARY
- A. Section includes:
  1. Wall Blokker by Commercial Acoustics soundproofing membrane.

1.2 RELATED DOCUMENTS
- A. Drawings and general provisions of the Contract including supplements and addendums.
- B. Applicable Specification Sections: Division 01 – General and Division 09 – Finishes.

1.3 REFERENCES
- A. International Building Code (IBC) 2012:
  1. Section 1207 – *Sound Transmission*
- B. ASTM Tests:
  1. E90 – *Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements.*
  2. E413 – *Classification for Rating Sound Insulation.*
- C. ASTM Specifications:
  1. C840 – *Standard Specification for Application and Finishing of Gypsum Board.*

1.4 SUBMITTALS
- A. For each product indicated:
  1. Product Data Sheet: manufacturer's specifications including laboratory test summary.
  2. Installation Instructions: detailed installation procedure including jobsite condition requirements, surface preparation requirements, and approved products.

1.5 DELIVERY, STORAGE, AND HANDLING
- A. All materials shall be delivered in original unopened packaging.
- B. Wall Blokker may be stored in a wet or dry environment and may be stored outdoors for up to three months on construction sites without special tarps or covering.
  1. If stored below freezing temperatures, material may require up to 48 hours of acclimation to regain pliability.
  2. Acclimate Wall Blokker for a minimum of 24 hours at temperatures 60 degrees (F) or greater to reduce material stiffness when handling.
  3. If material stiffens, it may be softened more rapidly using a heat gun.

1.6 PROJECT CONDITIONS
- A. Wall Blokker is typically installed after framing, insulation, and electrical are complete. Insulation should be installed in the wall cavity in addition for optimal performance.
- B. Ensure that all applicable inspections are completed prior to installation of Wall Blokker.
- C. Wall Blokker may be installed prior to "drying in" the building (prior to installation of windows and doors).
  1. Drywall should be installed within 2-4 weeks of Wall Blokker to prevent excessive wear.
     a. For longer delays, washers should be installed for securely fasten the material.

2. Drywall installation permanently attaches Wall Blokker to the stud.

PART 2 – PRODUCTS

2.1 WALL BLOKKER BY COMMERCIAL ACOUSTICS

A. Materials:
   1. Engineered sound reduction membrane.
   2. Flexible Ethylene Vinyl Acetate (EVA) product made from post-industrial recycled material.

B. Dimensions:
   1. Thickness: 1/8”
   2. Weight: 1 lb/sq.ft.
   3. Standard Sizes: 4’x25’ rolls; 4’x10’ and 4’x8’ sheets; custom length rolls available.
   4. Tolerances:
      a. Width: +/- 0.5”
      b. Length: +/- 1%
      c. Nominal Thickness: +/- 0.10”

C. Performance:
   1. Minimum STC = 25 (ASTM E90).
   2. UL Classified Assemblies:
      a. 300, 400, 500 Series
   3. Flammability Rating:
      a. Class 1 (ASTM E84 Rev. A)
      b. 1-Hour fire resistance wall rating (ASTM E119)
      c. 0.3 Thermal Resistance coefficient (ASTM C518)
   4. Environmental:
      a. Mold/Mildew resistant. No fungal/algae growth and no visible disfigurement (ASTM D3273 & ASTM G21).
      b. Impermeable air and moisture barrier.
      c. Non-PVC: no off-gassing.
      d. HIPPA Compliant.
      e. 100% recyclable at end of life.

PART 3 – EXECUTION

3.1 PREPARATION

A. Wall and/or stud assembly to receive Wall Blokker must be structurally sound prior to installation.
B. Wall must be clean and free of debris.
   1. Protrusions greater than 1/16” shall be scraped from the surface to avoid puncturing.
C. See Section 1.6 for additional project condition requirements.

3.2 INSTALLATION

A. Starting in one corner of the room, install Wall Blokker flush with the top of the top plate, and hang it vertically.
B. <u>Wood Studs:</u>
   1. Attach Wall Blokker to the top of the wood top plate using wide-crown ½” staples or pan head screws.

2. Fasten every 12” horizontally along the top plate.
3. Straighten Wall Blokker from the top down so that it is flush against the studs.
4. Attach Wall Blokker to the center of each vertical wood stud using ½” staples or pan head screws.
5. Fasten every 36” vertically along the center of each stud using an exterior fastener pattern, only fastening to the stud where the seam falls.

C. Metal Studs:
1. Attach Wall Blokker directly to the light gauge metal studs using drywall screws.
2. Fasten every 12” horizontally along the top.
   a. Wall Blokker installed on walls greater than 15’ in height shall be secured with washers along the top to prevent the fasteners from tearing the material.
   b. Fasteners shall be used on intermediate studs (in addition to exterior studs) every 12’ vertically.
3. Straighten Wall Blokker from the top down so that it is flush against the studs.
4. Attach Wall Blokker to the center of each vertical stud using drywall screws.
5. Fasten every 36” vertically along the center of each stud.

3.3 PROCEDURE

A. Install Wall Blokker as required on all walls.
B. Keep fasteners as flush as possible to prevent protrusion into the finished wallboard.
1. Fasteners shall not protrude more than 1/16” from Wall Blokker surface.
C. Do NOT overlap the seams of separate sheets.
1. Tightly butt the side of the next sheet of Wall Blokker to the edge of the existing attached sheet.
D. For seams that do not fall on a stud, tape with “Seam-Seal” or equivalent.
1. If seams fall on the stud with gaps greater than 1/8”, then taping is also required.
2. Ensure that there are no bubbles or wrinkles in the tape. Commercial tape alternatives include commercial duct tape.
3. The tape is semi-permanent and will be permanently sealed in position when drywall is installed. Drywall installation will seal Wall Blokker against the existing studs.
E. Cut Wall Blokker to fit around irregular objects and penetrations including outlets, switches, and junction boxes.
1. Gaps shall be less than 1/8”.
2. Gaps greater than 1/8” shall be sealed with acoustical or non-hardening caulk.
3. Gaps greater than 1/4” may be filled with backer rod or fiber batting.
4. Putty pads should be installed on the back of all electrical boxes.
F. Caulk the bottom of the floor plate at the floor line with acoustical sealant.
G. Install drywall per normal technique (ASTM C840).
1. Wall Blokker will be fastened permanently when the gypsum board is installed.
H. See Detailed Installation Instructions Figure 1 for diagrams.

CONTACT WITH QUESTIONS:
Commercial Acoustics, (888) 815-9691
1519 W Cypress St, Tampa, FL 33606

END OF SECTION

COMMERCIAL ACOUSTICS

Wall Blokker is a mass loaded, limp vinyl sound damping material designed for commercial, industrial, and residential applications to reduce sound transmission. It is used primarily behind finished wall or ceiling surfaces to block and damp noise through the entire sound spectrum.

Installation Instructions

Please check our website for the latest installation instructions:
http://commercial-acoustics.com/wp-content/uploads/2017/10/Wall-Blokker-Detailed-Installation.pdf

DO NOT USE WALL BLOKKER TO SURROUND OR ENCLOSE ANY LIGHT FIXTURES
CUT WALL BLOKKER BACK A MINIMUM OF 12" AWAY FROM ANY CANNED LIGHTS

Step 1
Preparation & Storage

Preparation:

1. Wall Blokker is typically installed after framing, insulation and electrical are complete. Insulation should be used in the wall cavity in addition for optimal performance.
2. Ensure that all applicable inspections are completed prior to installation of Wall Blokker
3. Wall Blokker may be installed prior to "drying in" the building (prior to installation of windows and doors)

Drywall should be installed within 2-4 weeks of Wall Blokker to prevent excessive wear. If longer delays are expected, washers should be installed to securely fasten Wall Blokker. Drywall installation permanently attaches the Wall Blokker to the stud.

Storage:

Wall Blokker may be stored in wet or dry environment, and may be stored outside for up to 3 months on construction sites without special tarps or covering. If stored below freezing temperatures, material may require 24-48 hours of acclimation to regain pliability.

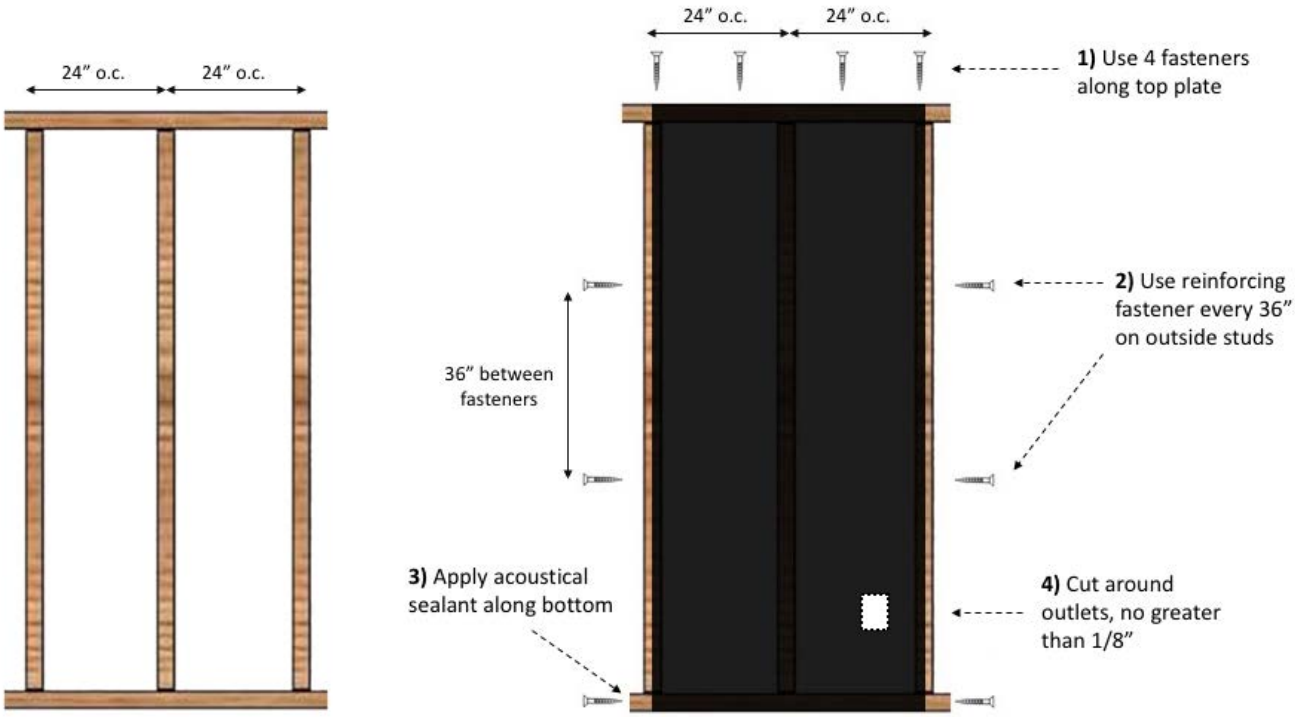
Step 2
Install Wall Blokker

1. Acclimate Wall Blokker for a minimum of 24 hours at temperatures 60°F or greater to reduce material stiffness when handling;
2. Starting in one corner of the room, install Wall Blokker flush with the top of top plate, and hang it vertically.
3. **Wood Studs:** Attach Wall Blokker directly to the top of the wood top plate using wide-crown ½" staples or pan head screws. A fastener every 12" horizontally is recommended along the top. Straighten Wall Blokker so that it is flush against the wood studs, and apply from the top down. Using staples or pan head screws, attach Wall Blokker to each stud in the center of the stud. Only 1 fastener is required every 36" for each vertical stud (refer to Diagram 1). Use an exterior fastener pattern, only fastening to the studs where the seam falls.
   - **NOTE:** Hammer Stapler may be best way to attach staples through Wall Blokker into stud.

   [See Figure 1 for details]
4. **Metal Studs:** Attach Wall Blokker directly to the metal stud using drywall screws. A screw

every 12" is recommended along the top. Straighten Wall Blokker so that it is flush against the metal studs. Using drywall screws, attach Wall Blokker to the stud in the middle of the stud. Only 1 fastener is required every 36" for each vertical stud. [See Figure 1 for details]

5. **NOTE: High Walls**
    - Wall Blokker on walls higher than 15' shall be secured with washers along the top to prevent the fasteners from pulling through the material
    - Fasteners should also be used on intermediate studs (in addition to exterior studs) every 12' vertically
6. Install Wall Blokker on all required walls;
7. Keep fasteners as flush as possible, since all protrusions will put a dimple into the finished wallboard. Fasteners shall not protrude more than 1/16" from Wall Blokker surface.
8. Tightly butt the side of the next sheet of Wall Blokker to the edge of the attached sheet. Do not overlap seams;
9. Tape all seams with "Seam-Seal" or equivalent. Ensure that there are no bubbles or wrinkles in the tape. The tape is semi-permanent, and will be permanently sealed in position when drywall is hung.
    - If seams fall on the stud, with no gaps >1/8", then no taping is required. Drywall installation will seal the Wall Blokker against the existing studs.
    - Commercial Tape Alternatives include commercial duct tape
10. Wall Blokker is easily cut to fit around irregular objects and electrical boxes. The material should be cut around outlets, switches, and junction boxes. Gaps shall be less 1/8". If greater, they shall be sealed with acoustical caulk. Gaps greater than ¼" may be filled with backer rod or fiber batting.
11. Putty pads should be installed on the back of all electrical boxes;
12. Caulk the bottom plate at the floor line with acoustical sealant;
13. Install drywall per normal technique (ASTM# C840 – Standard Specification for Application and Finishing of Gypsum Board). Wall Blokker will be fastened permanently when the gypsum board is hung.



[Figure 1]

6122 Benjamin Rd.| Tampa, Florida 33634 | 888.815.9691
info@commercial-acoustics.com | www.Commercial-Acoustics.com

TABLE 5-16 STC Ratings of Typical Wall Assemblies

Wall type: 2x4 wood studs, 16″ on-center (except as noted)	STC Rating (with insulation)	STC Rating (no insulation)	
Single layer 1/2″ drywall each side	38	35	
Single layer 1/2″ drywall one side, double layer 1/2″ drywall other side	40	38	
Double layer 1/2″ Type X drywall each side	45	39	
Resilient channel one side, single layer 1/2″ Type X drywall each side	47	39	
Staggered studs with 2x6 top and bottom plates. Single layer 1/2″ drywall each side	49	39	
Resilient channel with single layer 1/2″ drywall one side, double layer 1/2″ drywall other side	52	44	
Staggered studs 24″ o.c. with 2x6 top and bottom plates. Double layer 1/2″ drywall one side. Single layer 1/2″ Type X drywall other side	53	47	
Resilient channel one side with double layer 1/2″ Type X drywall each side	55	52	
Staggered studs 24″ o.c. with 2x6 top and bottom plates. Double layer 1/2″ Type X drywall each side	55	52	